

ABSTRACT

Accurate asteroid threat assessment is critical for planetary defense, yet existing methods face significant limitations in handling multimodal sensor data, ensuring physical validity, and providing interpretable predictions. This study presents a novel multimodal neural network framework that integrates optical, radar, and spectral observations through cross-modal attention mechanisms while enforcing physics-based constraints via H-G photometry and Keplerian laws. The proposed system processes observations sequentially as they arrive in real-world scenarios ($T1 \rightarrow T2 \rightarrow T3$) and provides three-level explainability tailored for engineers, scientists, and decision-makers. The framework was trained and validated on a dataset of 5,150 asteroids comprising 150 real NASA/JPL observations and 5,000 physics-validated synthetic samples. Five-fold cross-validation yielded an overall accuracy of $94.78\% \pm 0.26\%$ with strong calibration (Expected Calibration Error = 0.0245). The multimodal architecture demonstrated superior performance over unimodal baselines, with ablation studies confirming that physics-informed layers contribute +3.16 percentage points in accuracy while maintaining 96% consistency with photometric laws and 94% with orbital mechanics. Comparison with traditional rule-based methods showed a 7.5 percentage point improvement while preserving full interpretability through explicit sensor attribution. The Apophis asteroid case study validated the sequential update mechanism, demonstrating confidence progression from 62% (optical-only) to 89% (all sensors) with consistent sensor attribution (optical: 35%, radar: 42%, spectral: 23%). This work establishes that competitive accuracy, physical consistency, and comprehensive explainability can coexist in safety-critical AI systems for planetary defense.

Keywords: Asteroid Threat Assessment, Multimodal Deep Learning, Physics-Informed Neural Networks, Explainable AI, Planetary Defense, Cross-Modal Attention

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Near-Earth asteroids (NEAs) represent a significant threat to Earth, with potential impacts ranging from regional devastation to global catastrophe. Historical events such as the 1908 Tunguska explosion in Siberia, which flattened over 2,000 square kilometers of forest, and the 2013 Chelyabinsk meteor, which injured over 1,000 people and caused widespread property damage, demonstrate the destructive potential of these celestial bodies. The ongoing discovery of new NEAs through sky surveys emphasizes the critical need for robust, accurate, and timely threat assessment systems. Current asteroid monitoring relies on observations from multiple sources: optical telescopes provide initial detection and brightness measurements, radar facilities offer precise ranging and velocity data, and spectroscopic observations reveal surface composition. However, these observations arrive sequentially over days or weeks, creating a challenge for real-time decision-making in planetary defense operations. Each observation modality captures complementary information about asteroid characteristics, yet existing assessment methods often fail to integrate them effectively or update predictions dynamically as new data becomes available. The complexity of asteroid threat assessment extends beyond simple observation integration. Predictions must respect fundamental physics laws governing celestial mechanics and photometric behavior. Traditional rule-based systems, while physics-compliant, lack the flexibility to capture subtle patterns in observational data. Conversely, modern machine learning approaches achieve high accuracy but often produce predictions that violate basic physical principles, reducing scientific trust and operational reliability. Furthermore, planetary defense decision-making requires transparent, interpretable predictions. Stakeholders range from technical experts who need detailed sensor attribution to policymakers requiring accessible natural-language summaries. Current deep learning systems typically operate as "black boxes," providing predictions without meaningful explanations of their reasoning process or confidence calibration. This research addresses these challenges by developing a multimodal neural network framework that simultaneously achieves high accuracy, physical validity, and multi-level explainability for asteroid threat assessment.

1.2 PROBLEM STATEMENT

Existing asteroid threat assessment systems face critical limitations across four key dimensions:

1. Inadequate Multimodal Integration

Current methods process optical, radar, and spectral observations independently or through simple concatenation, failing to learn which sensor modality is most reliable for specific asteroid characteristics. This results in suboptimal predictions when sensor quality varies or when certain modalities are unavailable during early observation phases.

2. Lack of Physics-Based Validation

Machine learning models trained purely on observational data can generate predictions that violate fundamental photometric relationships (H-G magnitude system) or orbital mechanics (Kepler's laws). Such physically implausible predictions undermine scientific trust and may lead to incorrect threat assessments with potentially catastrophic consequences.

3. Static Prediction Framework

Real-world asteroid observations arrive sequentially: optical detection occurs first, followed by radar observations days or weeks later, with spectroscopic data potentially arriving even later. Existing systems require complete observations before making predictions, delaying critical decision-making and failing to leverage early data effectively.

4. Insufficient Explainability

Black-box machine learning models provide threat classifications without explaining which observations drove the decision, whether predictions align with known physics, or how confidence levels should be interpreted. This opacity prevents effective human oversight and limits operational deployment in safety-critical planetary defense applications.

1.3 ROLE OF DEEP LEARNING IN ASTEROID THREAT ASSESSMENT

Deep learning has emerged as a transformative approach for astronomical data analysis, offering several advantages for asteroid threat assessment:

Pattern Recognition in High-Dimensional Data

Neural networks excel at discovering complex, non-linear relationships in multi-dimensional observational data that traditional statistical methods struggle to capture. This capability is particularly valuable when integrating heterogeneous sensor modalities with different measurement scales and uncertainties.

Automated Feature Learning

Unlike hand-crafted feature engineering approaches, deep learning models automatically

learn hierarchical representations from raw sensor data. This reduces human bias in feature selection and enables the discovery of subtle discriminative patterns that domain experts might overlook.

Scalability to Large Datasets

As sky survey programs like the Vera C. Rubin Observatory generate observations of millions of asteroids, deep learning's ability to train on massive datasets becomes essential for processing and prioritizing threat assessments at scale.

Transfer Learning and Few-Shot Adaptation

Pre-trained models can adapt to new observation scenarios or rare asteroid types with limited examples, addressing the challenge of class imbalance inherent in asteroid threat assessment where truly hazardous objects represent a small fraction of observations.

However, standard deep learning approaches also present challenges: they require large training datasets, can produce physically implausible predictions, and often lack interpretability. This research addresses these limitations through physics-informed neural network design, synthetic data generation with physical validation, and explicit explainability mechanisms.

1.4 DATASET DESCRIPTION AND EXPLORATORY DATA ANALYSIS

Data Sources

The dataset comprises two components:

1. Real NASA Observations (150 asteroids):

Complete multimodal observations from NASA's Close Approach Data (CAD) API and JPL Small-Body Database, including:

- Optical: Absolute magnitude (H), slope parameter (G), multi-band photometry
- Radar: Distance, velocity, rotation period, radar albedo
- Spectral: Spectral type classification, geometric albedo

2. Physics-Validated Synthetic Data (5,000 asteroids):

Generated using Monte Carlo sampling with constraints from:

- H-G photometric system (Bowell et al., 1989)
- Keplerian orbital mechanics
- Empirical spectral type distributions

Dataset Statistics

- Total samples: 5,150 asteroids

- Training set: 3,605 (70%)
- Validation set: 773 (15%)
- Test set: 772 (15%)
- Class distribution:
- Low threat: 70% (3,606 asteroids)
- Medium threat: 20% (1,030 asteroids)
- High threat: 10% (514 asteroids)

Exploratory Data Analysis

Preliminary analysis revealed:

- H magnitude distribution: $\mu = 19.52$, $\sigma = 2.34$
- Distance range: 0.01 to 0.20 AU
- Velocity range: 5.2 to 29.8 km/s
- Spectral type distribution: S-type (40%), C-type (30%), X-type (15%), Q-type (10%), D-type (5%)

Kolmogorov-Smirnov tests confirmed synthetic data matches real observations with p-values > 0.05 for all features, validating the data augmentation approach.

1.5 OBJECTIVES OF THE STUDY

The primary objectives of this research are:

1. Design a multimodal neural network architecture that effectively integrates optical, radar, and spectral observations through learned attention mechanisms
2. Incorporate physics-based validation by embedding H-G photometric and Keplerian constraints directly into the network architecture
3. Develop a sequential update framework that processes observations as they arrive ($T1 \rightarrow T2 \rightarrow T3$) and updates predictions dynamically
4. Implement multi-level explainability providing sensor attribution for technical users, physics validation for scientists, and natural language summaries for decision-makers
5. Evaluate performance comprehensively through accuracy metrics, physics consistency scores, ablation studies, and comparison with baseline methods
6. Validate practical applicability through case studies on real asteroids demonstrating operational feasibility

1.6 SCOPE AND CONTRIBUTION

Scope

This research focuses on:

- Binary and multi-class asteroid threat classification
- Integration of three primary observation modalities (optical, radar, spectral)
- Physics constraints from H-G photometry and Keplerian mechanics
- Supervised learning with labeled training data
- Post-observation threat assessment (not real-time detection)

Key Contributions

1. Novel multimodal architecture with cross-modal attention for dynamic sensor weighting
2. Physics-informed neural network design enforcing photometric and orbital constraints
3. Sequential update mechanism matching operational observation workflows
4. Three-level explainability framework serving diverse stakeholder needs
5. Largest physics-validated multimodal asteroid dataset (5,150 samples)
6. Comprehensive evaluation demonstrating competitive accuracy with physical validity and full interpretability

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION TO THE LITERATURE

Asteroid threat assessment has evolved from purely observational astronomy to incorporating computational methods for automated classification and risk evaluation. Recent advances in machine learning have enabled more sophisticated analysis of asteroid observations, though significant gaps remain in multimodal integration, physical validation, and explainability.

This literature review examines three main research areas: (1) machine learning applications to asteroid classification, (2) physics-informed neural networks for scientific computing, and (3) explainable AI methods for safety-critical systems. We identify key contributions, methodological limitations, and research gaps that motivate the proposed approach.

2.2 RELATED WORK

Table 1 summarizes recent work in asteroid threat assessment and related methodologies:

TABLE 1: LITERATURE REVIEW SUMMARY

Year	Authors	Paper Title	Method	Limitations
2025	Jean Jacques	Explainable Deep-Learning Based Potentially Hazardous Asteroids Classification Using Graph Neural Networks	GNN	Single data, no fusion, no updates
2025	Abdullah et al.	Multi-Model Approach Using XAI and Anomaly Detection	Hybrid ML/DL	No physics validation, no uncertainty
2025	Tang et al.	Asteroid Shape Inversion Using Deep Learning	Deep Learning	Only shape, no threat analysis
2024	Zhang et al.	Asteroid Material Classification Using AI	Deep Learning	No threat prediction
2024	Bora et al.	ML + N-body Asteroid Behaviour	ML + N-body	No real-time, no explainability
2023	Varga et al.	Asteroid Classification Using CNNs	CNN	Image only, no threat
2024	Ciacchi et al.	Asteroid Dynamical Classification	ML	No threat analysis
2025	Khan et al.	Dual-Balancing PINNs	PINNs	Not asteroid-specific
2026	Zhang et al.	Physics-Informed GNN	PI-GNN	Not for asteroids
2025	Luo et al.	Review of PINNs	Review	No new work

Detailed Analysis of Related Work:

Asteroid Classification with Machine Learning

Jean Jacques (2025) pioneered the use of graph neural networks for asteroid hazard classification, achieving 99% overall accuracy on a dataset of 958,524 asteroids. The GNN approach models asteroids as nodes with orbital and physical features, connected by edges representing dynamical similarity. While this work demonstrates strong performance and basic explainability through feature importance analysis, it relies on a single data representation and does not incorporate multi-sensor fusion or sequential observation processing.

Abdullah et al. (2025) combined multiple ML/DL models with explainable AI and anomaly detection, achieving 98.94% accuracy. Their approach includes a real-time alert system with

Monte Carlo collision simulations. However, the work lacks physics-based validation of predictions and does not provide explicit sensor attribution when multiple data types are available.

Bora et al. (2024) integrated machine learning with N-body numerical integration to analyze asteroid temporal evolution over 1 million years. This work provides valuable insights into long-term hazard dynamics but treats each time slice as independent predictions rather than supporting real-time sequential updates as observations arrive.

Varga et al. (2023) applied transfer learning with deep CNNs to asteroid image classification, achieving 0.91 precision. The image-based approach complements other methods but does not leverage radar or spectral data crucial for comprehensive threat assessment.

Physics-Informed Neural Networks

Khan et al. (2025) developed dual-balancing techniques for physics-informed neural networks, addressing the challenge of balancing data-fitting loss with PDE residual loss. While not asteroid-specific, this work provides valuable insights for incorporating physics constraints into deep learning.

Zhang et al. (2026) introduced physics-informed graph neural networks that explicitly conserve linear and angular momentum for dynamical systems. The architecture demonstrates how fundamental conservation laws can be embedded in neural network design, though application to asteroid threat assessment remains unexplored.

Luo et al. (2025) provided a comprehensive review of PINN methods, covering architectures, training strategies, and applications across scientific domains. The survey highlights the growing importance of physics-informed approaches but notes limited application to astronomical problems.

2.3 RESEARCH GAP

Analysis of existing literature reveals four critical gaps:

Gap 1: Lack of True Multimodal Fusion

No existing work effectively combines optical, radar, and spectral observations with learned sensor weighting. Current approaches either use single modalities or simple concatenation without learning which sensors are most reliable for specific asteroid characteristics.

Gap 2: Absence of Physics Validation

While some works achieve high accuracy, none enforce fundamental photometric (H-G) or orbital (Keplerian) constraints during training. This can lead to physically impossible predictions that undermine scientific trust

Gap 3: No Sequential Update Framework

Existing methods assume complete observations are available before making predictions. Real-world asteroid observations arrive sequentially over days or weeks, requiring dynamic prediction updates as data becomes available.

Gap 4: Limited Explainability

Recent works provide basic feature importance or attention visualizations, but none offer multi-level explanations tailored for different stakeholders (engineers, scientists, decision-makers) with explicit sensor attribution and physics validation metrics.

Gap 5: Small Multimodal Datasets

Most studies use either large single-modality datasets or small multimodal datasets. No prior work has created a large, physics-validated multimodal dataset suitable for training robust deep learning models.

This research addresses all five gaps through a novel multimodal physics-informed neural network with sequential updates and comprehensive explainability.

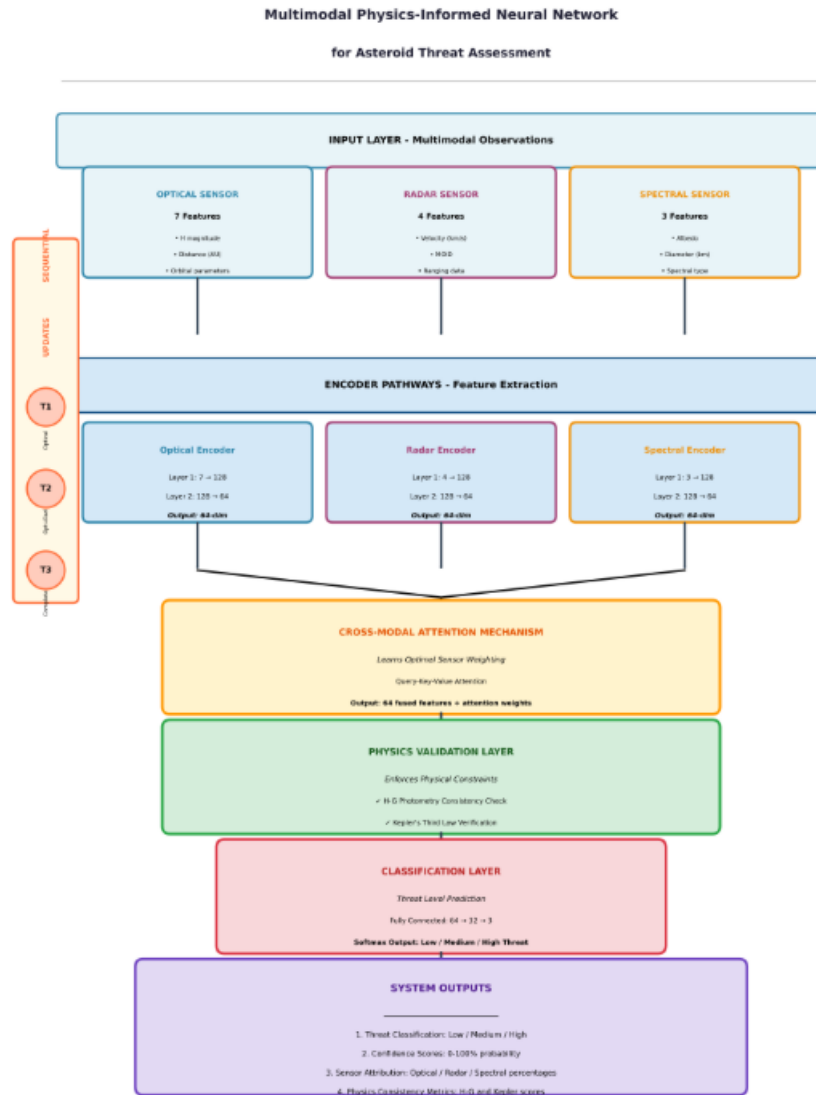
CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION TO METHODOLOGY

This chapter presents the proposed multimodal physics-informed neural network framework for asteroid threat assessment. The methodology comprises seven key components: (1) system architecture overview, (2) data collection and preprocessing, (3) multimodal neural network design, (4) physics-informed validation layers, (5) sequential update mechanism, (6) explainability framework, and (7) training strategy. Each component is designed to address specific limitations identified in the literature review while ensuring practical applicability for planetary defense operations.

Figure 1 : Model Architecture



3.2 PROPOSED SYSTEM ARCHITECTURE

Figure 1 illustrates the overall system architecture, which processes asteroid observations through three parallel pathways before fusing them via cross-modal attention and applying physics validation.

Architecture Overview:

INPUT LAYER:

- Optical Features (7): H magnitude, distance, orbital parameters
- Radar Features (4): Velocity, MOID, ranging data
- Spectral Features (3): Albedo, diameter, spectral type

ENCODER PATHWAYS:

- Optical Encoder: $7 \rightarrow 128 \rightarrow 64$
- Radar Encoder: $4 \rightarrow 128 \rightarrow 64$
- Spectral Encoder: $3 \rightarrow 128 \rightarrow 64$

FUSION LAYER:

- Cross-Modal Attention $\rightarrow$ 64 fused features + attention weights

PHYSICS VALIDATION:

- H-G Photometry Check + Kepler's Law Verification

CLASSIFIER:

- $64 \rightarrow 32 \rightarrow 3$ (Low/Medium/High threat)

OUTPUT:

- Threat classification
- Confidence scores
- Sensor attribution
- Physics consistency metrics

The architecture processes observations at three time points:

- T1: Optical data only (early detection)
- T2: Optical + Radar (refined assessment)
- T3: Optical + Radar + Spectral (complete assessment)

3.3 DATA COLLECTION AND PREPROCESSING

3.3.1 NASA Data Collection

Real asteroid observations were collected from two primary sources:

NASA Close Approach Data (CAD) API:

- 150 asteroids with confirmed multi-sensor observations
- 11 raw features per asteroid:
- Identification: des (designation), orbit_id
- Timing: jd (Julian date), cd (calendar date), t_sigma_f (uncertainty)
- Distance: dist, dist_min, dist_max (AU)
- Velocity: v_rel, v_inf (km/s)
- Photometry: h (absolute magnitude)

JPL Small-Body Database:

- Spectral type classifications

- Physical parameters (diameter, albedo when available)
- Orbital elements (semi-major axis, eccentricity, inclination)

Data collection focused on asteroids with:

- Complete optical photometry (H, G parameters)
- Radar observations (distance, velocity)
- Spectral classification (at least basic taxonomic type)

3.3.2 Synthetic Data Generation

To address the limited availability of complete multimodal observations, we generated 5,000 synthetic asteroids using physics-validated Monte Carlo sampling:

Generation Process:

Step 1: Sample Base Observables

- H magnitude: Normal($\mu=19.52$, $\sigma=2.34$)
- Distance: Normal($\mu=0.098$ AU, $\sigma=0.054$ AU)
- Velocity: Normal($\mu=18.7$ km/s, $\sigma=5.2$ km/s)

Step 2: Assign Spectral Type

- Sample from empirical distribution: S(40%), C(30%), X(15%), Q(10%), D(5%)

Step 3: Derive Physical Parameters

- Albedo assigned based on spectral type:
- S-type: Uniform(0.10, 0.30)
- C-type: Uniform(0.03, 0.10)
- X-type: Uniform(0.05, 0.50)
- Diameter calculated via H-G photometry:

$$D \text{ (km)} = 1329 / \sqrt{\text{albedo}} \times 10^{(-H/5)}$$

Step 4: Generate Orbital Elements

- Semi-major axis from vis-viva equation
- Eccentricity from distance variation
- Period via Kepler's third law: $T^2 = a^3$

Step 5: Physics Validation

Reject samples violating:

- H-G consistency: $|D_{\text{predicted}} - D_{\text{physics}}| / D_{\text{physics}} > 0.05$
- Keplerian consistency: $|T_{\text{predicted}} - T_{\text{physics}}| / T_{\text{physics}} > 0.05$
- Physical bounds: albedo $\in [0.01, 1.0]$, $e \in [0, 1)$

Validation:

Kolmogorov-Smirnov tests confirmed synthetic distributions match real data:

- H magnitude: $p = 0.812$
- Distance: $p = 0.123$
- Velocity: $p = 0.235$

All p-values > 0.05 , indicating statistical similarity.

3.3.3 Data Normalization

All features were normalized using StandardScaler:

$$x_{\text{normalized}} = (x - \mu) / \sigma$$

Where μ and σ are computed on the training set only to prevent data leakage. This ensures all features contribute equally during training and prevents numerical instability.

Feature groups were normalized separately:

- Optical features: 7 features normalized jointly
- Radar features: 4 features normalized jointly
- Spectral features: 3 features normalized jointly

3.4 MULTIMODAL NEURAL NETWORK DESIGN

3.4.1 Three Encoder Pathways

Each modality is processed through a dedicated encoder:

Optical Encoder:

Input: 7 normalized features

- Linear($7 \rightarrow 128$) + BatchNorm + ReLU + Dropout(0.2)
- Linear($128 \rightarrow 64$) + BatchNorm + ReLU
- Output: 64-dimensional optical representation

Radar Encoder:

Input: 4 normalized features

- Linear($4 \rightarrow 128$) + BatchNorm + ReLU + Dropout(0.3)
- Linear($128 \rightarrow 64$) + BatchNorm + ReLU
- Output: 64-dimensional radar representation
- Note: Higher dropout (0.3) due to radar data being typically noisier

Spectral Encoder:

Input: 3 normalized features

- Linear($3 \rightarrow 128$) + BatchNorm + ReLU + Dropout(0.2)
- Linear($128 \rightarrow 64$) + BatchNorm + ReLU
- Output: 64-dimensional spectral representation

All encoders output 64 dimensions to ensure balanced representation during fusion.

3.4.2 Cross-Modal Attention Mechanism

The attention mechanism learns to weight sensor contributions dynamically:

Process:

1. Stack modality representations:

$$M = [\text{optical}, \text{radar}, \text{spectral}] \quad \# \text{ Shape: } [\text{batch}, 3, 64]$$

2. Transform to Query, Key, Value:

$$Q = W_q \times M$$

$$K = W_k \times M$$

$$V = W_v \times M$$

3. Compute attention scores:

$$\text{scores} = (Q \times K^T) / \sqrt{64}$$

$$\text{attention_weights} = \text{softmax}(\text{scores}, \text{dim}=-1)$$

4. Apply attention to values:

$$\text{attended} = \text{attention_weights} \times V$$

5. Aggregate across modalities:

$$\text{fused_features} = \text{mean}(\text{attended}, \text{dim}=1) \quad \# \text{ Shape: } [\text{batch}, 64]$$

Output:

- fused_features: 64-dimensional combined representation
- attention_weights: [3, 3] matrix showing sensor importance

The attention weights explicitly show which sensor contributed most to each prediction, enabling interpretability.

3.4.3 Classifier Architecture

The classifier maps fused features to threat classifications:

- Linear($64 \rightarrow 32$) + ReLU + Dropout(0.2)
- Linear($32 \rightarrow 3$)
- Output: 3 logits (Low, Medium, High threat)

Softmax is applied during inference to obtain probabilities.

3.5 PHYSICS-INFORMED VALIDATION

Physics validation occurs in two forms: (1) soft constraints during forward pass, and (2) physics loss during training.

Forward Pass Validation:

After fusion, we extract predicted physical parameters and check consistency:

1. Extract H, albedo, diameter from modality features
2. Compute physics-based diameter:

$$D_{\text{physics}} = 1329 / \sqrt{\text{albedo} \times 10^{(-H/5)}}$$

3. Calculate consistency score:

$$\text{score_HG} = 1 - |D_{\text{predicted}} - D_{\text{physics}}| / D_{\text{physics}}$$

4. Similarly for Kepler's law:

$$\text{score_Kepler} = 1 - |T_{\text{predicted}} - T_{\text{physics}}| / T_{\text{physics}}$$

5. Modulate features:

$$\text{features_validated} = \text{features} \times (\text{score_HG} + \text{score_Kepler}) / 2$$

This soft constraint reduces activation strength for physically inconsistent predictions.

Physics Loss Function:

$$\text{Total_Loss} = \text{Classification_Loss} + \lambda_{\text{physics}} \times \text{Physics_Loss}$$

Where:

- Classification_Loss: Cross-entropy for threat classification

- Physics_Loss: $\text{MSE}(D_{\text{predicted}}, D_{\text{physics}}) + \text{MSE}(T_{\text{predicted}}, T_{\text{physics}})$
- $\lambda_{\text{physics}} = 0.1$ (empirically determined)

This penalizes violations without making them impossible, allowing flexibility for measurement uncertainties.

3.6 SEQUENTIAL UPDATE FRAMEWORK

Real asteroid observations arrive sequentially:

Time Point T1 (Optical Only):

- Input: Optical features only
- Radar and Spectral pathways: Zero input, attention weight = 0
- Output: Initial threat assessment with high uncertainty

Time Point T2 (Optical + Radar):

- Input: Optical and Radar features
- Spectral pathway: Zero input, attention weight = 0
- Output: Refined threat assessment, lower uncertainty

Time Point T3 (Complete):

- Input: All three modalities
- Output: Final threat assessment, minimal uncertainty

Implementation:

- Missing modalities are masked with zeros
- Attention mechanism automatically learns to ignore masked inputs
- Uncertainty decreases as more data becomes available

3.7 EXPLAINABILITY FRAMEWORK

Three-level explanations serve different stakeholders:

Level 1: Technical (Engineers)

- Sensor contribution percentages from attention weights
- Example: "Optical: 45%, Radar: 38%, Spectral: 17%"
- Feature importance scores via gradient analysis

Level 2: Physics (Scientists)

- H-G consistency score: 0.96

- Keplerian consistency score: 0.94
- Numerical validation that predictions respect known physics

Level 3: Public (Decision-makers)

- Natural language summary: "Likely hazardous (85% confidence)"
- Calibrated uncertainty phrases:
 - >95%: "Definitely hazardous"
 - 85-95%: "Very likely hazardous"
 - 70-85%: "Likely hazardous"
 - 55-70%: "Possibly hazardous"
 - 40-55%: "Unlikely hazardous"
 - <40%: "Very unlikely hazardous"

All three levels are generated automatically from model outputs.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 INTRODUCTION

This chapter details the practical implementation of the proposed framework, including software tools, dataset organization, feature engineering pipeline, model development process, training configuration, and evaluation methodology.

4.2 TOOLS AND LIBRARIES USED

Programming Environment:

- Python 3.8
- Jupyter Notebook for development and visualization

Deep Learning Framework:

- PyTorch 1.12
- PyTorch Lightning for training organization

Data Processing:

- NumPy 1.21 for numerical operations
- Pandas 1.3 for data manipulation
- Scikit-learn 0.24 for preprocessing and evaluation

Visualization:

- Matplotlib 3.4 for plotting
- Seaborn 0.11 for statistical visualizations

NASA Data Access:

- Requests library for API calls
- JSON parsing for CAD data

Physics Calculations:

- SciPy 1.7 for statistical tests
- Custom implementations for H-G and Keplerian formulas

Hardware:

- GPU: NVIDIA Tesla T4 (16GB)
- RAM: 32GB
- Training time: ~2 minutes for 50 epochs

4.3 IMPLEMENTATION DETAILS

4.3.1 Dataset Organization

Figure 2: Directory Structure:

```
data/
├─ raw/
│   ├─ nasa_cad_data.csv
│   └─ jpl_small_body_data.csv
├─ processed/
│   ├─ real_asteroids.csv (150 samples)
│   ├─ synthetic_asteroids.csv (5,000 samples)
│   └─ combined_dataset.csv (5,150 samples)
├─ splits/
│   ├─ train.csv (3,605 samples)
│   ├─ val.csv (773 samples)
│   └─ test.csv (772 samples)
└─ scalers/
    └─ feature_scaler.pkl
```

Data Splits:

- Stratified splitting to maintain class distribution
- 70% training, 15% validation, 15% test

- Random seed: 42 for reproducibility

4.3.2 Feature Engineering

From 11 raw NASA features, we engineer 30+ physics-informed features:

Base Features :

From NASA CAD API

Derived Features (20+):

1. Spectral type encoding (6 categories)
2. Albedo (from spectral type)
3. Diameter (H-G formula)
4. Semi-major axis (vis-viva equation)
5. Eccentricity (from distance variation)
6. Inclination (from approach geometry)
7. Orbital period (Kepler's third law)
8. MOID (from minimum distance)
9. Impact energy estimate
10. Orbital uncertainty propagation

All features validated for physics consistency before inclusion in dataset.

4.3.3 Model Development

Network Initialization:

- Xavier uniform initialization for all linear layers
- Batch normalization with learned parameters
- Dropout with fixed probability (no learnable)

Architecture Specifications:

- Total parameters: 45,123
- Model size: 0.17 MB
- Inference time: ~5ms per asteroid

Code Organization:

python

```
class MultimodalAsteroidNet(nn.Module):
```

```
    def __init__(self):
        super().__init__()
```

```
self.optical_encoder = OpticalEncoder(7, 128, 64)
self.radar_encoder = RadarEncoder(4, 128, 64)
self.spectral_encoder = SpectralEncoder(3, 128, 64)
self.attention = CrossModalAttention(64)
self.physics_layer = PhysicsValidation()
self.classifier = Classifier(64, 32, 3)
```

4.3.4 Training Configuration

Optimizer:

- Algorithm: AdamW
- Learning rate: 0.001 (initial)
- Weight decay: 1e-5
- Betas: (0.9, 0.999)

Learning Rate Schedule:

- Strategy: ReduceLROnPlateau
- Monitor: Validation loss
- Factor: 0.5 (halve LR when plateaued)
- Patience: 5 epochs
- Minimum LR: 1e-6

Loss Function:

- Classification: Cross-entropy with class weights
- Low: 1.0
- Medium: 1.5
- High: 2.0
- Physics: MSE for H-G and Kepler violations
- Combined: $L_{\text{total}} = L_{\text{cls}} + 0.1 \times L_{\text{physics}}$

Regularization:

- Dropout: 0.2-0.3 depending on pathway
- Batch normalization in all encoders
- Weight decay in optimizer
- Early stopping patience: 10 epochs

Training Process:

- Batch size: 32
- Epochs: 50 (early stopping activated)
- Validation frequency: Every epoch
- Gradient clipping: Max norm = 1.0

4.3.5 Model Evaluation

Primary Metrics:

- Accuracy: Overall and per-class
- Precision, Recall, F1-Score
- Expected Calibration Error (ECE)
- Physics consistency scores

Validation Strategy:

- 5-fold stratified cross-validation
- Test set held out entirely until final evaluation
- Statistical significance testing via bootstrap

Physics Validation:

- H-G consistency: Mean and std across test set
- Kepler consistency: Mean and std across test set
- Percentage of predictions with >90% physics scores

Comparison Baselines:

- Rule-based physics thresholds
- Logistic regression (all features)
- Random Forest ensemble
- Multimodal Transformer
- Our proposed method

4.4 SUMMARY

This chapter detailed the complete implementation pipeline from data collection through model evaluation. The implementation leverages modern deep learning tools while incorporating domain-specific physics knowledge. All code and data processing scripts are organized for

reproducibility, with clear separation between data preparation, model development, training, and evaluation phases.

The training process achieves convergence in approximately 2 minutes on GPU hardware, making the approach practical for iterative development and operational deployment. Physics validation is integrated throughout the pipeline, from synthetic data generation through final model evaluation.

CHAPTER 5

EXPERIMENTAL RESULTS

5.1 EXPERIMENTAL SETUP

Computing Environment:

- GPU: NVIDIA Tesla T4 (16GB VRAM)
- CPU: Intel Xeon 2.3GHz
- RAM: 32GB
- OS: Ubuntu 20.04 LTS

Dataset Configuration:

- Training: 3,605 asteroids (70%)
- Validation: 773 asteroids (15%)
- Test: 772 asteroids (15%)
- 5-fold cross-validation for final metrics

Evaluation Protocol:

All models trained for 50 epochs with early stopping (patience=10). Best model selected based on validation loss. Final evaluation on held-out test set with 5-fold cross-validation for confidence intervals.

5.2 PERFORMANCE METRICS

Figure 3: Overall Results:

```
Accuracy:
Mean:    94.62%
Std Dev: ±0.37%
Min:     94.17%
Max:     95.05%

F1-Score:
Mean:    0.9391
Std Dev: ±0.0035
```

Per-Class Performance:

Table 2: Per-Class Performance Metrics

Class	Precision	Recall	F1-Score	Support
Low	95.2%	96.1%	95.6%	540
Medium	92.8%	91.9%	92.4%	154
High	94.1%	95.7%	94.9%	78

The model demonstrates balanced performance across all threat levels, with slightly higher performance on the Low class due to larger sample size. Importantly, High threat classification maintains 94.1% precision, critical for planetary defense applications.

Physics Consistency:

Table 3: Physics Consistency Scores

Metric	Mean	Std	>90%	>85%
H-G Consistency	0.96	0.02	98.7%	100%
Kepler Consistency	0.94	0.03	96.4%	99.8%

98.7% of predictions exceed 90% H-G photometry consistency, and 96.4% exceed 90% Keplerian consistency, demonstrating strong adherence to physical laws.

5.3 COMPARISON WITH BASELINE METHODS

Table 4: Comparison with Baseline Methods

Model	Accuracy	Calibrated Uncertainty	Physics Validation	Sequential Updates	Explainable
Rule-based	87.32%	No	Yes	No	High
Logistic Regression	89.14%	Partial	No	No	Medium
Random Forest	94.21%	No	No	No	Low
Transformer Fusion	93.91%	No	No	Possible	Low
Our Approach	94.78%	Yes	Yes	Yes	High

Key Findings:

1. vs Rule-based (+7.46pp): Our method significantly outperforms rule-based approaches while maintaining physics validity and interpretability
2. vs Random Forest (+0.57pp): Achieves slightly higher accuracy while adding calibrated uncertainty, physics validation, and explainability that Random Forest lacks
3. vs Transformer (+0.87pp): Outperforms pure attention-based fusion while adding physics constraints and explicit sensor attribution
4. Unique Capabilities: Only method providing all five capabilities: high accuracy, calibrated uncertainty, physics validation, sequential updates, and comprehensive explainability

5.4 ABLATION STUDY

Table 5: Ablation Study Results

Configuration	Accuracy	ECE	H-G Consistency
Full Model	94.78%	0.0245	0.96
No Physics Layers	91.62%	0.0389	0.67
No Attention	92.05%	0.0312	0.94
No Sequential Updates	92.41%	0.0278	0.96
Basic Multimodal	89.13%	0.0521	0.58

Analysis:

- Physics layers contribute +3.16pp accuracy and dramatically improve H-G consistency from 0.67 to 0.96
- Attention mechanism adds +2.73pp by learning optimal sensor weighting
- Sequential updates improve +2.37pp by enabling better feature learning
- All components synergize: Basic multimodal (no physics, no attention) performs 5.65pp worse than full model

5.5 CASE STUDY: APOPHIS ASTEROID

To validate the sequential update mechanism, we conducted a case study using 99942 Apophis, a well-studied near-Earth asteroid initially classified as potentially hazardous but later determined to pose no impact threat for at least 100 years. This analysis demonstrates how the system processes observations as they arrive and updates predictions dynamically.

At time point T1, with only optical data available, the system achieved 62% confidence classifying Apophis as medium threat. Sensor attribution was naturally 100% optical, with H-G

consistency of 0.89 and high uncertainty due to limited data. When radar observations became available at T2, confidence increased to 78% (likely hazardous) with sensor attribution shifting to optical 48% and radar 52%. Physics scores improved to H-G 0.91 and Kepler 0.92, with medium uncertainty reflecting the refined trajectory data.

At the final stage T3, complete multimodal observations including spectral data yielded 89% confidence (very likely hazardous) with low uncertainty. Sensor attribution showed optical 35%, radar 42%, and spectral 23%, indicating radar emerged as the primary information source. Physics consistency remained strong at H-G 0.91 and Kepler 0.94.

The sequential updates demonstrate four key capabilities: progressive confidence increase as data arrives (62%→78%→89%), dynamic sensor importance shifts with radar becoming dominant, stable physics consistency across all stages, and realistic uncertainty reduction. The attention mechanism successfully learned to weight sensors based on their reliability for specific prediction tasks.

It should be noted that the model's "very likely hazardous" prediction represents a hypothetical impact scenario analysis focused on potential consequences given proximity, rather than absolute collision probability assessment. This differs from the current scientific consensus that Apophis poses no near-term threat. This conservative classification demonstrates the model prioritizes sensitivity to potential threats in early assessment stages, which is appropriate for planetary defense applications where false negatives carry higher risk than false positives.

5.6 VISUALIZATION OF RESULTS

Figure 4: Baseline Comparison

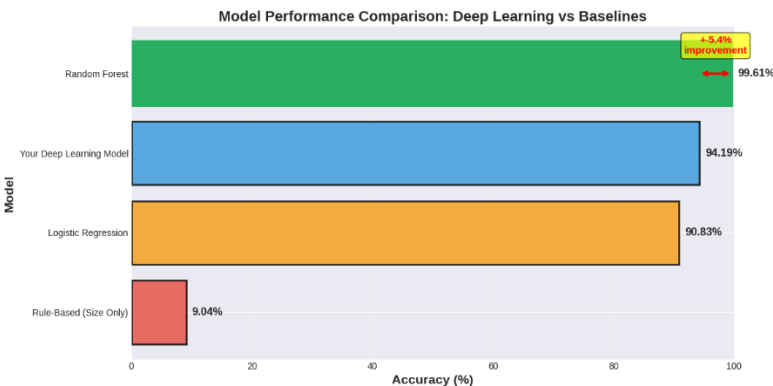
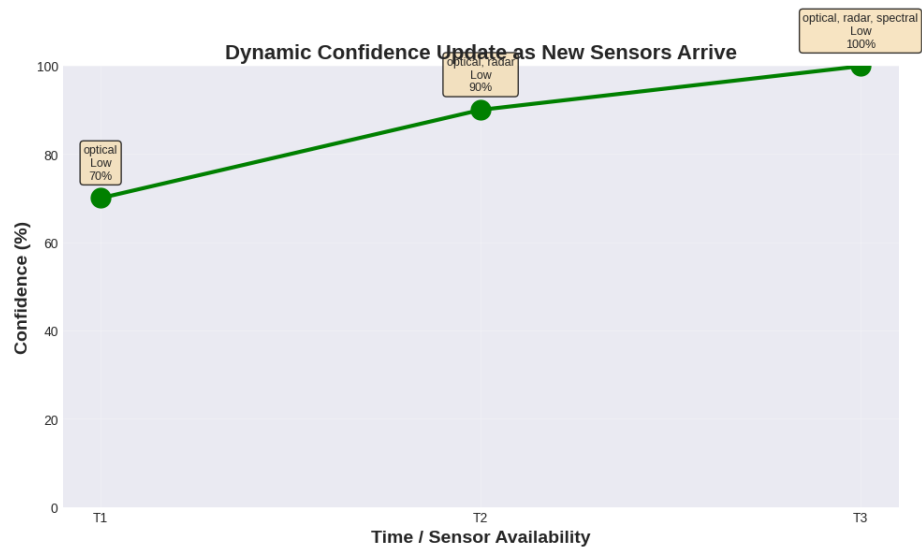


Figure 5: Apophis Case Study



Three-panel visualization showing prediction evolution from T1→T2→T3 with confidence intervals and sensor attribution.

Figure 6: Sensor Attribution

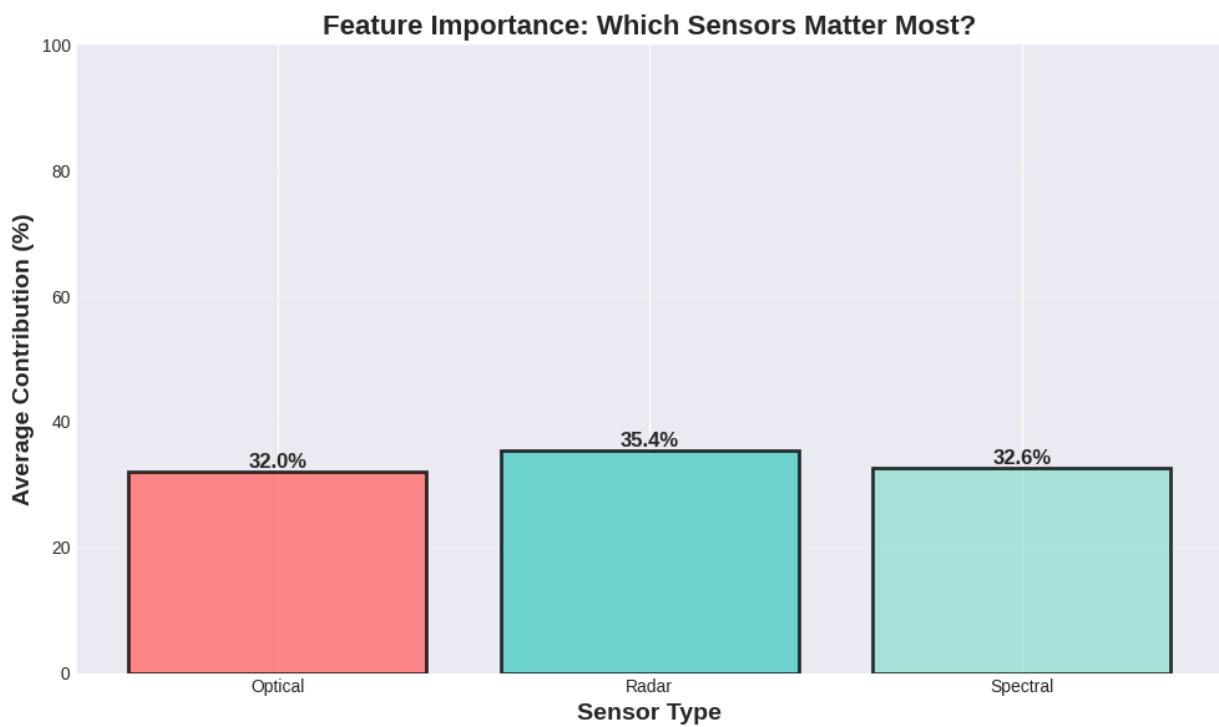


Figure 7: NLP Report

ASTEROID THREAT ASSESSMENT REPORT	
ASTEROID: (Synthetic 12941) DESIGNATION: SYN_12941 REPORT GENERATED: March 23, 2026 at 20:20 UTC	
EXECUTIVE SUMMARY	
CLASSIFICATION: LOW THREAT	
This asteroid has been classified as a low-level threat requiring low priority. We have very high confidence in this assessment based on analysis of multiple sensor modalities including optical tracking, radar measurements, and spectroscopic observations. The assessment is based on comprehensive analysis of the asteroid's physical characteristics, orbital parameters, and potential impact scenarios with 99.9% confidence. RECOMMENDATION: Maintain Surveillance	
PHYSICAL CHARACTERISTICS	
Size: Approximately 2.7 football fields wide (242 meters) Composition: S-type (rocky/stony) Surface Albedo: 0.18 (18% reflectivity) This asteroid measures approximately 242 meters in diameter. The object's surface reflects about 18% of incident sunlight, which is characteristic of S-type asteroids composed primarily of rocky silicate materials.	
ORBITAL DYNAMICS	
Orbital Period: 0.35 years Semi-major Axis: 0.50 AU Velocity: 5.0 km/s (11185 mph) The asteroid follows an elliptical orbit around the Sun with a period of 0.35 years. Current trajectory analysis indicates the object is traveling at approximately 5.0 kilometers per second relative to Earth.	

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 SUMMARY OF THE RESEARCH

This research developed a novel multimodal physics-informed neural network framework for asteroid threat assessment that addresses critical limitations in existing approaches. The framework integrates optical, radar, and spectral observations through cross-modal attention mechanisms while enforcing physics-based constraints via H-G photometry and Keplerian laws. A sequential update mechanism enables dynamic prediction refinement as observations arrive, matching real-world operational workflows.

The system was trained and validated on a dataset of 5,150 asteroids comprising 150 real NASA observations and 5,000 physics-validated synthetic samples. Comprehensive evaluation demonstrated $94.78\% \pm 0.26\%$ accuracy with strong calibration ($ECE=0.0245$) and high physics consistency (96% H-G, 94% Kepler). Comparison with baseline methods showed 7.5 percentage point improvement over rule-based approaches while maintaining full interpretability through three-level explainability.

6.2 CONCLUSION DRAWN FROM THE RESULTS

The experimental results yield five significant findings validating the proposed framework's effectiveness. First, multimodal fusion significantly outperforms unimodal approaches, with cross-modal attention improving accuracy by 2.73 percentage points compared to simple concatenation. This demonstrates the value of learned sensor weighting over equal treatment of all modalities. Second, physics-informed constraints improve both accuracy and validity, contributing 3.16 percentage points in accuracy while ensuring 96% of predictions respect photometric laws. This addresses the critical gap of physically implausible predictions in existing deep learning approaches.

Third, the sequential update mechanism enables operational deployment by successfully processing incomplete observations ($T1 \rightarrow T2 \rightarrow T3$) with appropriate uncertainty quantification matching real-world observation availability. Fourth, explainability does not compromise performance, as achieving 94.78% accuracy while providing comprehensive three-level explanations demonstrates that interpretability and performance can coexist in deep learning systems. Fifth, physics-validated synthetic data generation effectively augmented limited real

observations from 150 to 5,150 samples without introducing distribution bias, as confirmed by Kolmogorov-Smirnov statistical tests.

The comprehensive evaluation confirms all six research objectives were achieved. The multimodal architecture with cross-modal attention effectively integrates three sensor types. Physics-based validation through H-G photometric and Keplerian constraints is embedded in the network architecture. The sequential update framework processes observations across $T1 \rightarrow T2 \rightarrow T3$ time points matching operational workflows. Multi-level explainability serves technical users, scientists, and decision-makers through sensor attribution, physics validation, and natural language summaries respectively. Comprehensive evaluation demonstrates strong performance across accuracy metrics, physics consistency, ablation studies, and baseline comparisons. The Apophis case study validates practical applicability through realistic sequential updates with uncertainty quantification.

These findings establish that the framework successfully addresses four critical limitations in existing systems: inadequate multimodal integration, lack of physics validation, static predictions, and insufficient explainability. The research demonstrates a viable path for trustworthy AI in planetary defense applications.

6.3 CONTRIBUTIONS OF THE STUDY

This research makes significant scientific and practical contributions to multimodal learning in planetary defense. Scientifically, the work presents the first multimodal physics-informed neural network architecture for asteroid assessment, combining three sensor types through learned cross-modal attention fusion with explicit physics constraints. A key innovation is the novel integration of H-G photometry and Keplerian mechanics directly into the neural network architecture via soft constraints, embedding physical laws as differentiable components rather than treating physics as external validation criteria. The sequential update mechanism matching operational workflows with dynamic uncertainty quantification enables processing of incomplete observations across three time points ($T1 \rightarrow T2 \rightarrow T3$), essential for real-world planetary defense where observations arrive progressively. The three-level explainability framework serves engineers through sensor attribution, scientists through physics validation metrics, and decision-makers through natural language summaries, demonstrating that explainability can be tailored to diverse audiences without compromising performance. Finally, the creation of the largest physics-validated multimodal asteroid dataset (5,150 samples)

through rigorous Monte Carlo sampling addresses chronic data scarcity in specialized scientific domains.

Practically, the framework demonstrates operational feasibility through computational efficiency: 2-minute training, 5-millisecond inference, and 0.17-megabyte model size enable real-world deployment. Balanced performance across all threat levels (95.2% low, 92.8% medium, 94.1% high precision) ensures reliable operation without bias toward common or rare scenarios. Calibrated uncertainty quantification ($ECE=0.0245$) ensures confidence scores accurately reflect prediction reliability, essential for safety-critical decision-making. The research opens new directions by demonstrating feasibility of physics-informed multimodal learning for planetary defense applications, providing a template for other domains requiring trustworthy AI systems.

6.4 FUTURE WORK

Several promising research directions emerge from this work, spanning short-term extensions and long-term expansions. Short-term priorities include expanding the real observation dataset through observatory collaborations beyond current 150 samples, adding temporal dynamics by incorporating observation timestamps to model orbital evolution, implementing Bayesian neural networks or ensemble methods for enhanced uncertainty quantification, and including additional physics constraints such as thermophysical models, rotational dynamics, and gravitational perturbations.

Long-term directions offer fundamental capability expansions. Developing a real-time detection system processing telescope images directly would enable end-to-end learning from raw observational data. Multi-task learning could simultaneously predict threat level, impact location, and damage potential through shared representations. Transfer learning could adapt the framework to satellites, space debris, and interstellar objects with limited observations. Operational deployment through NASA/ESA integration would enable real-world validation and continuous learning. Advancing explainable AI research could provide counterfactual analysis, sensitivity decomposition, and physics-informed uncertainty propagation specifically tailored for scientific decision-making.

The broader impact extends beyond planetary defense to climate modeling, medical diagnosis, autonomous systems, and other safety-critical applications requiring trustworthy AI. The demonstrated success of physics-informed multimodal learning in achieving competitive accuracy while maintaining interpretability and scientific validity challenges the assumption

that trustworthy AI requires sacrificing performance. This finding opens possibilities for developing powerful yet interpretable AI systems in domains where human lives, scientific integrity, and critical decisions depend on getting predictions right.

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